

JEGADEESH D

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SUMMARY

AI Engineer (Integrated M.Tech, Sept 2025) building end-to-end applied AI systems — agent runtimes, hybrid RAG pipelines, and ML services — with Python, FastAPI, TypeScript/React 18, and PostgreSQL/SQLite. Ships deployed, containerized, and test-covered systems (281+ automated pytest tests across 4 Dockerized projects on GCP Cloud Run), with a focus on deterministic API contracts and quantitative LLM/retrieval evaluation.

CORE COMPETENCIES & SKILLS

AI & Agentic Systems — Agent Runtimes, Tool-Calling, Hybrid RAG (Dense + BM25), Reciprocal Rank Fusion (RRF), ChromaDB (Vector DB), Sentence Transformers, LLM APIs (OpenRouter Multi-Model Gateway), scikit-learn, PyTorch

Backend & Systems — Python (AsyncIO), FastAPI, REST APIs, Docker, GCP Cloud Run, PostgreSQL, SQLite (FTS5), Linux, Git, Uvicorn, Pydantic-Settings

Testing & Reliability — pytest (281+ Automated Tests), pytest-asyncio, Evaluator Calibration Benchmarks, Unit & Integration Testing, CI/CD

Full-Stack & Frontend — TypeScript, React 18, Vite, Schema Validation, Responsive UI, Streamlit (Internal Ops Cockpits)

PROJECTS & INITIATIVES

Financial Regulatory Intelligence Engine — Hybrid Dense + Sparse RAG *Python, FastAPI, ChromaDB, BM25, Docker, pytest*

- Architected an enterprise hybrid retrieval pipeline over 12,075 semantic chunks from a 12-document corpus of regulatory circulars and corporate annual reports spanning BFSI and other sectors (RBI, SEBI, IRDAI, HDFC, ICICI, TCS, RIL, Tata Consumer), fusing dense MiniLM embeddings with BM25 sparse keyword ranking via Reciprocal Rank Fusion (RRF; $k = 60$).
- Benchmarked retrieval offline against a 10-question labeled evaluation set, lifting Mean Reciprocal Rank (MRR) to 0.550 and Top-5 Hit Rate to 70.0% (7/10) independently of LLM generation.
- Deployed containerized FastAPI service with Portkey gateway failover, confidence gating, and page-level citations; validated by 88 automated pytest tests. [\[Live\]](#) [\[GitHub\]](#)

GoalOS — Local-First Executive Agent Runtime

React 18, TypeScript, FastAPI, SQLite FTS5, ChromaDB

- Engineered a local-first agent runtime with React 18 + Vite frontend and FastAPI backend, exposing 6 schema-validated tools across 4 domain namespaces with a 100% (7/7) execution-reliability and unauthorized-tool-rejection benchmark pass rate.
- Implemented a 5-factor composite memory retrieval engine combining SQLite FTS5 full-text lexical search with ChromaDB cosine similarity, exponential recency decay, and importance weighting for zero-latency recall.
- Built automatic failover to deterministic local heuristic engines during offline operation or API rate limits; validated by 130 automated pytest tests. [\[Live\]](#) [\[GitHub\]](#)

SuperKalam — Agentic UPSC Mains Evaluator

Python, FastAPI, SQLite, ChromaDB, OpenRouter, Pydantic v2

- Designed an autonomous multi-agent pipeline (Retrieval → Evaluator → Feedback) scoring student answers against 40 curated Civil Services-style previous year questions across 4 rubric dimensions.
- Enforced strict Pydantic v2 JSON schema contracts on all LLM evaluator output, with automatic retry-and-repair on malformed responses and HTTP 429 circuit breaking.
- Generated native-script feedback in Hindi and Tamil; validated by 27 automated pytest tests. [\[Live\]](#) [\[GitHub\]](#)

Customer Support Analytics — Multi-Task ML & Agentic Triage

Python, FastAPI, scikit-learn, PostgreSQL, Docker

- Architected a two-tier triage system over 200K tickets: Tier 1 classical models handle 73.6% of routine traffic without escalation at sub-50ms median latency (\$0 API cost; 82.1% priority-classification accuracy, R-squared = 0.719 SLA resolution-time regression, zero data leakage).
- Implemented Tier 2 agentic escalation activating only for low confidence (< 0.67), severe resolution (> 185 h), or a high-risk Enterprise segment, generating structured JSON with customer frustration diagnostics and auto-drafted replies.
- Deployed FastAPI REST endpoints (POST /predict_*, POST /triage_agent) with PostgreSQL orchestration; validated by 36 automated pytest tests. [\[Live\]](#) [\[GitHub\]](#)

EDUCATION

MTech (Integrated) in Software Engineering — Vellore Institute of Technology (VIT), Chennai Jul 2020 – Sep 2025
CGPA: 7.89 / 10.0 | **Coursework:** Data Structures & Algorithms, Software Architecture, Distributed Systems, Database Systems, Machine Learning